05 Sequences and Series

1. DEFINITION:

Sequence:

A succession of terms a_1, a_2, a_3, a_4..... formed according to some rule or law.

Examples are : 1, 4, 9, 16, 25

$-1, 1, -1, 1$,.....

$\frac{x}{1!}, \frac{x^2}{2!}, \frac{x^3}{3!}, \frac{x^4}{4!}$,.....

A finite sequence has a finite (i.e. limited) number of terms, as in the first example above. An infinite sequence has an unlimited number of terms, i.e. there is no last term, as in the second and third examples.

Series:

The indicated sum of the terms of a sequence. In the case of a finite sequence $a_1, a_2, a_3, \ldots, a_n$ the corresponding series is $a_1 + a_2 + a_3 + \ldots + a_n = \sum_{k=1}^{n} a_k$. This series has a finite or limited number of terms and is called a finite series.

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2. ARITHMETIC PROGRESSION (A.P.):

A.P. is a sequence whose terms differ by a fixed number. This fixed number is called the common difference. If a is the first term & d the common difference, then A.P. can be written as

$a, a + d, a + 2d, \ldots, a + (n - 1)\, d$, ...

(a) n^{th} term of AP $T_n = a + (n-1)d$, where d = $t_n - t_{n-1}$

(b) The sum of the first n terms : $S_n = \frac{n}{2}[a + \ell] = \frac{n}{2}[2a + (n-1)d]$

where ℓ is n^{th} term.

Note :

(i) n^{th} term of an A.P. is of the form $An + B$ i.e. a linear expression in 'n', in such a case the coefficient of n is the common difference of the A.P. i.e. A.

(ii) Sum of first 'n' terms of an A.P. is of the form $An^2 + Bn$ i.e. a quadratic expression in 'n', in such case the common difference is twice the coefficient of n^2. i.e. $2A$

(iii) Also n^{th} term $T_n = S_n - S_{n-1}$

3. PROPERTIES OF A.P.:

(a) If each term of an A.P. is increased, decreased, multiplied or divided by the some nonzero number, then the resulting sequence is also an A.P.

(b) Three numbers in A.P. : $a - d, a, a + d$

Four numbers in A.P. : $a - 3d, a - d, a + d, a + 3d$

Five numbers in A.P. : $a - 2d, a - d, a, a + d, a + 2d$

Six numbers in A.P. : $a - 5d, a - 3d, a - d, a + d, a + 3d, a + 5d$ etc.

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(c) The common difference can be zero, positive or negative.

(d) k^{th} term from the last = $(n - k + 1)^{th}$ term from the beginning (If total number of terms = n).

(e) The sum of the two terms of an AP equidistant from the beginning & end is constant and equal to the sum of first & last terms. $\Rightarrow T_k + T_{n-k+1}$ = constant = $a + \ell$.

(f) Any term of an AP (except the first) is equal to half the sum of terms which are equidistant from it. $a_n = (1/2)(a_{n-k} + a_{n+k}), k < n$
For $k = 1, a_n = (1/2)(a_{n-1} + a_{n+1})$; For $k = 2, a_n = (1/2)(a_{n-2} + a_{n+2})$ and so on.

(g) If a, b, c are in AP, then $2b = a + c$.

Illustration 1

If $(x + 1), 3x$ and $(4x + 2)$ are first three terms of an A.P. then its 5th term is -

(A) 14 (B) 19 (C) 24 (D) 28

Ans. (C)

Solution:

$(x + 1), 3x, (4x + 2)$ are in AP

$\Rightarrow 3x - (x + 1) = (4x + 2) - 3x \Rightarrow x = 3$

$\therefore a = 4, d = 9 - 4 = 5 \Rightarrow T_5 = 4 + (4)5 = 24$

Illustration 2

The sum of first four terms of an A.P. is 56 and the sum of its last four terms is 112. If its first term is 11 then find the number of terms in the A.P.

Solution:

$a + a + d + a + 2d + a + 3d = 56$

$\Rightarrow 4a + 6d = 56$

$\Rightarrow 44 + 6d = 56$ (as $a = 11$)

$\Rightarrow 6d = 12$ hence $d = 2$

Let total number of terms = n

Now sum of last four terms.

$\Rightarrow a + (n - 1)d + a + (n - 2)d + a + (n - 3)d + a + (n - 4)d = 112$

$\Rightarrow 4a + (4n - 10)d = 112 \Rightarrow 44 + (4n - 10)2 = 112$

$\Rightarrow 4n - 10 = 34$

$\Rightarrow n = 11$ **Ans.**

Illustration 3

The sum of first n terms of two A.Ps. are in ratio $\frac{7n+1}{4n+27}$. Find the ratio of their 11th terms.

Solution:

Let a_1 and a_2 be the first terms and d_1 and d_2 be the common differences of two A.P.s respectively then

$$\frac{\frac{n}{2}[2a_1 + (n-1)d_1]}{\frac{n}{2}[2a_2 + (n-1)d_2]} = \frac{7n+1}{4n+27} \Rightarrow \frac{a_1 + \left(\frac{n-1}{2}\right)d_1}{a_2 + \left(\frac{n-1}{2}\right)d_2} = \frac{7n+1}{4n+27}$$

For ratio of 11th terms

$$\Rightarrow \frac{n-1}{2} = 10 \Rightarrow n = 21$$

so ratio of 11th terms is $\frac{7(21)+1}{4(21)+27} = \frac{148}{111} = \frac{4}{3}$ **Ans.**

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Illustration 4

Given the sequence of numbers $x_1, x_2, x_3, \ldots\ldots x_{2013}$

which satisfy $\frac{x_1}{x_1+1} = \frac{x_2}{x_2+3} = \frac{x_3}{x_3+5} = \ldots\ldots = \frac{x_{2013}}{x_{2013}+4025}$, nature of the sequence is

(A) A.P. (B) G.P. (C) H.P. (D) A.G.P.

Ans. (A)

Solution:

$\frac{x_1}{x_1+1} = \frac{x_2}{x_2+3} = \frac{x_3}{x_3+5} = \ldots\ldots = \frac{x_{2013}}{x_{2013}+4025} = \frac{1}{\lambda}$

$\Rightarrow x_1 = \frac{1}{\lambda-1}, x_2 = \frac{3}{\lambda-1}, x_3 = \frac{5}{\lambda-1}, \ldots\ldots\ldots, x_{2013} = \frac{4025}{\lambda-1}$

$\Rightarrow x_1, x_2, x_3, \ldots\ldots\ldots, x_{2013}$ are in A.P. with common difference $= \frac{2}{\lambda-1} = d$

Illustration 5

If 1, 2, 3 ... are first terms; 1, 3, 5 .... are common differences and S_1, S_2, S_3 .... are sums of n terms of given p AP's; then $S_1 + S_2 + S_3 + \ldots + S_p$ is equal to

(A) $\frac{np(np+1)}{2}$ (B) $\frac{n(np+1)}{2}$ (C) $\frac{np(p+1)}{2}$ (D) $\frac{np(np-1)}{2}$

Ans. (A)

Solution:

$S_1 + S_2 + S_3 + \ldots\ldots\ldots + S_p$

$\Rightarrow S_1 = \frac{n}{2}[2.1 + (n-1)\,1]$

$S_2 = \frac{n}{2}[2.2 + (n-1)\,3]$

$S_3 = \frac{n}{2}[2.3 + (n-1)\,5]$

$\vdots \qquad \vdots$

$S_p = \frac{n}{2}[2.p + (n-1)\,(2p-1)]$

So $S_1 + S_2 + \ldots\ldots + S_p = \frac{n}{2}[2(1 + 2 + \ldots\ldots p) + (n-1)\,(1 + 3 + 5 + \ldots\ldots (2p-1))]$

$= \frac{n}{2}\left[2.\frac{p\ (p+1)}{2} + (n-1)p^2\right] = \frac{np}{2}(np+1)$ **Ans.**

Illustration 6

If the sum of the first m terms of an A.P. is equal to the sum of either the next n terms or the next p terms, then prove that $(m+n)\left(\frac{1}{m} - \frac{1}{p}\right) = (m+p)\left(\frac{1}{m} - \frac{1}{n}\right)$.

Solution:

Let a denotes first term and d common difference of the A.P. then a_k, the kth term of the A.P.

$a_k = a + (k-1)\,d$; $a_1 + a_2 + a_3 + \ldots\ldots + a_m = a_{m+1} + a_{m+2} + \ldots\ldots + a_{m+n}$

$2S_m = S_{m+n}$; $2 \cdot \frac{m}{2}[2a + (m-1)\,d] = \frac{m+n}{2}[2a + (m+n-1)\,d]$

$\frac{2m}{m+n} = \frac{2a + (m+n-1)\ d}{2a + (m-1)\ d} = 1 + \frac{nd}{2a + (m-1)\ d}$

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$\Rightarrow \frac{nd}{2a+(m-1)\ d} = \frac{m-n}{m+n}$ ...(i)

similarly $\frac{pd}{2a+(m-1)d} = \frac{m-p}{m+p}$ ...(ii)

divide (i) by (ii)

$\frac{n}{p} = \frac{m-n}{m+n} \cdot \frac{m+p}{m-p} \Rightarrow \frac{(m+n)(m-p)}{p} = \frac{(m+p)(m-n)}{n} \Rightarrow \frac{(m+n)(m-p)}{mp} = \frac{(m+p)(m-n)}{mn}$

$(m+n)\left(\frac{1}{m}-\frac{1}{p}\right) = (m+p)\left(\frac{1}{m}-\frac{1}{n}\right)$ Hence Proved

Illustration 7

Four numbers are in A.P. If their sum is 20 and the sum of their squares is 120, then the middle terms are -

(A) 2, 4 (B) 4, 6 (C) 6, 8 (D) 8, 10

Ans. (B)

Solution:

Let the numbers are $a - 3d, a - d, a + d, a + 3d$

given, $a - 3d + a - d + a + d + a + 3d = 20 \Rightarrow 4a = 20 \Rightarrow a = 5$

and $(a - 3d)^2 + (a - d)^2 + (a + d)^2 + (a + 3d)^2 = 120 \Rightarrow 4a^2 + 20d^2 = 120$

$\Rightarrow 4 \times 5^2 + 20d^2 = 120 \Rightarrow d^2 = 1 \Rightarrow d = \pm 1$

Hence numbers are 2, 4, 6, 8 or 8, 6, 4, 2

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Illustration 8

If $a_1, a_2, a_3, ..., a_n$ are in A.P. where $a_i > 0$ for all i, show that :

$$\frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+.....+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}=\frac{(n-1)}{\sqrt{a_1}+\sqrt{a_n}}$$

Solution:

$$\text{L.H.S.} = \frac{1}{\sqrt{a_1}+\sqrt{a_2}}+\frac{1}{\sqrt{a_2}+\sqrt{a_3}}+.....+\frac{1}{\sqrt{a_{n-1}}+\sqrt{a_n}}$$

$$= \frac{1}{\sqrt{a_2}+\sqrt{a_1}}+\frac{1}{\sqrt{a_3}+\sqrt{a_2}}+.....+\frac{1}{\sqrt{a_n}+\sqrt{a_{n-1}}}$$

$$= \frac{\sqrt{a_2}-\sqrt{a_1}}{(a_2-a_1)}+\frac{\sqrt{a_3}-\sqrt{a_2}}{(a_3-a_2)}+.....+\frac{\sqrt{a_n}-\sqrt{a_{n-1}}}{a_n-a_{n-1}}$$

Let 'd' is the common difference of this A.P.

then $a_2 - a_1 = a_3 - a_2 = ... = a_n - a_{n-1} = d$

Now L.H.S.

$$= \frac{1}{d}\left\{\sqrt{a_2}-\sqrt{a_1}+\sqrt{a_3}-\sqrt{a_2}+.....+\sqrt{a_{n-1}}-\sqrt{a_{n-2}}+\sqrt{a_n}-\sqrt{a_{n-1}}\right\} = \frac{1}{d}\left\{\sqrt{a_n}-\sqrt{a_1}\right\}$$

$$= \frac{a_n-a_1}{d\left(\sqrt{a_n}+\sqrt{a_1}\right)} = \frac{a_1+(n-1)d-a_1}{d\left(\sqrt{a_n}+\sqrt{a_1}\right)} = \frac{1}{d}\frac{(n-1)d}{(\sqrt{a_n}+\sqrt{a_1})} = \frac{n-1}{\sqrt{a_n}+\sqrt{a_1}} = R.H.S.$$

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Illustration 9

The series of natural numbers is divided into groups (1), (2, 3, 4), (5, 6, 7, 8, 9) ..... and so on. Show that the sum of the numbers in n^{th} group is $n^3 + (n - 1)^3$

Solution:

The groups are (1), (2, 3, 4), (5, 6, 7, 8, 9) .....

The number of terms in the groups are 1, 3, 5.....

$\therefore$ The number of terms in the nth group = $(2n - 1)$

the last term of the nth group is n^2

If we count from last term common difference should be –1

So, the sum of numbers in the nth group = $\left(\frac{2n-1}{2}\right)\left\{2n^2+(2n-2)(-1)\right\}$

= $(2n - 1)(n^2 - n + 1) = 2n^3 - 3n^2 + 3n - 1 = n^3 + (n - 1)^3$

4. GEOMETRIC PROGRESSION (G.P.):

G.P. is a sequence of non-zero numbers in which, each of the succeeding term is equal to the preceding term multiplied by a constant. Thus, in a GP the ratio of successive terms is constant. This constant factor is called the COMMON RATIO of the sequence & is obtained by dividing any term by the immediately previous term. Therefore a, ar, ar^2, ar^3, ar^4 , ... is a GP with 'a' as the first term & 'r' as common ratio.

(a) n^{th} term ; $T_n = a\,r^{n-1}$

(b) Sum of the first n terms; $S_n = \frac{a(r^n-1)}{r-1}$, if $r \neq 1$

(c) Sum of infinite G.P. , $S_\infty = \frac{a}{1-r}$; $0 < |r| < 1$

5. PROPERTIES OF GP:

(a) If each term of a G.P. be multiplied or divided by some non-zero quantity, then the resulting sequence is also a G.P.

(b) Three consecutive terms of a GP: $a/r,\ a, ar$;
Four consecutive terms of a GP: $a/r^3,\ a/r, ar, ar^3$ & so on.

(c) If a, b, c are in G.P. then $b^2 = ac$.

(d) In a G.P, the product of two terms which are equidistant from the first and the last term is constant and is equal to the product of first and last term. $\Rightarrow T_k \,.\, T_{n-k+1}$ = constant = $a.\ell$

(e) If each term of a G.P. be raised to the same power, then resulting sequence is also a G.P.

(f) In a G.P., $T_r^2 = T_{r-k}.T_{r+k}$, $k < r$, $r \neq 1$

(g) If the terms of a given G.P. are chosen at regular intervals, then the new sequence is also a G.P.

(h) If $a_1,\ a_2,\ a_3 ....a_n$ is a G.P. of positive terms, then log a_1, log a_2,.....log a_n are an A.P. and vice-versa.

(i) If a_1, a_2, a_3..... and b_1, b_2, b_3..... are two G.P.'s then a_1b_1, a_2b_2, a_3b_3..... & $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}$..... is also in G.P.

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Illustration 10

If a, b, c, d and p are distinct real numbers such that $\left(a^2+b^2+c^2\right)p^2-2p\left(ab+bc+cd\right)+\left(b^2+c^2+d^2\right)\le 0$ then a, b, c, d are in

(A) A.P. (B) G.P. (C) H.P. (D) None of these

Ans. (B)

Solution:

Here, the given condition $\left(a^2+b^2+c^2\right)p^2-2p\left(ab+bc+cd\right)+b^2+c^2+d^2\le 0$

$\Rightarrow \left(ap-b\right)^2+\left(bp-c\right)^2+\left(cp-d\right)^2\le 0$

$\because$ a square can not be negative

$\therefore \quad ap-b=0, bp-c=0, cp-d=0 \Rightarrow p=\frac{b}{a}=\frac{c}{b}=\frac{d}{c}\Rightarrow$ a, b, c, d are in G.P.

Illustration 11

If positive real numbers a, b, c are in G.P., then the equations $ax^2+2bx+c=0$ and $dx^2+2ex+f=0$ have a common root if $\frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in -

(A) A.P. (B) G.P. (C) H.P. (D) None of these

Ans. (A)

Solution:

a, b, c are in G.P $\Rightarrow b^2=ac$

Now the equation $ax^2+2bx+c=0$ can be rewritten as $ax^2+2\sqrt{ac}x+c=0$

$\Rightarrow \left(\sqrt{a}x+\sqrt{c}\right)^2=0\Rightarrow x=-\sqrt{\frac{c}{a}}, -\sqrt{\frac{c}{a}}$

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If the two given equations have a common root, then this root must be $-\sqrt{\frac{c}{a}}$.

Thus $d\frac{c}{a}-2e\sqrt{\frac{c}{a}}+f=0 \Rightarrow \frac{d}{a}+\frac{f}{c}=\frac{2e}{c}\sqrt{\frac{c}{a}}=\frac{2e}{\sqrt{ac}}=\frac{2e}{b} \Rightarrow \frac{d}{a}, \frac{e}{b}, \frac{f}{c}$ are in A.P.

Illustration 12

A number consists of three digits which are in G.P. the sum of the right hand and left hand digits exceeds twice the middle digit by 1 and the sum of the left hand and middle digits is two third of the sum of the middle and right hand digits. Find the numbers.

Solution:

Let the three digits be a, ar and ar^2 then number is

$100a + 10ar + ar^2$ ...(i)

Given, $a + ar^2 = 2ar + 1$

or $a(r^2 - 2r + 1) = 1$

or $a(r - 1)^2 = 1$ ...(ii)

Also given $a + ar = \frac{2}{3}(ar + ar^2)$

$\Rightarrow 3 + 3r = 2r + 2r^2 \Rightarrow 2r^2 - r - 3 = 0 \Rightarrow (r + 1)(2r - 3) = 0$

$\therefore r = -1, 3/2$

for $r = -1$, $a = \frac{1}{(r-1)^2}=\frac{1}{4}\notin I \quad \therefore r \neq -1$

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for $r = 3/2$, $a = \dfrac{1}{\left(\frac{3}{2}-1\right)^2} = 4$ {from (ii)}

From (i), number is $400 + 10.4.\ \frac{3}{2} + 4.\frac{9}{4} = 469$ **Ans.**

Illustration 13

Find the value of $0.32\overline{58}$

Solution:

Let $R = 0.32\overline{58} \Rightarrow R = 0.32585858....$ ...(i)

Here number of figures which are not recurring is 2 and number of figures which are recurring is also 2.

then $100\,R = 32.585858.....$ ...(ii)

and $10000\,R = 3258.5858.....$ ...(iii)

Subtracting (ii) from (iii) , we get

$9900\,R = 3226 \Rightarrow R = \dfrac{1613}{4950}$

Aliter Method: $R = .32 + .0058 + .000058 + .00000058 + ...$

$$= .32 + \frac{58}{10^4}\left(1 + \frac{1}{10^2} + \frac{1}{10^4} + ...\infty\right) = .32 + \frac{58}{10^4}\left(\frac{1}{1-\frac{1}{100}}\right)$$

$$= \frac{32}{100} + \frac{58}{9900} = \frac{3168+58}{9900} = \frac{3226}{9900} = \frac{1613}{4950}$$

Illustration 14

If the sum of n terms of a G.P. (with common ratio r) beginning with the pth term is k times the sum of an equal number of terms of the same series beginning with the qth term, then the value of k is:

(A) $r^{p/q}$ (B) $r^{q/p}$ (C) r^{p-q} (D) r^{p+q}

Solution:

Ans. (C)

$\dfrac{ar^{p-1}(r^n-1)}{r-1} = k\ \dfrac{ar^{q-1}(r^n-1)}{r-1}$.; $r^{p-1} = k\,.\,r^{q-1}$; $k = r^{p-q}$

Illustration 15

The sides of a right triangle form a G.P. The tangent of the smallest angle is

(A) $\sqrt{\dfrac{\sqrt{5}+1}{2}}$ (B) $\sqrt{\dfrac{\sqrt{5}-1}{2}}$ (C) $\sqrt{\dfrac{2}{\sqrt{5}+1}}$ (D) $\sqrt{\dfrac{2}{\sqrt{5}-1}}$

Ans. (B,C)

Solution:

Case - I

$r > 1$

$a^2 + a^2r^2 = a^2r^4 \Rightarrow r^4 - r^2 - 1 = 0$

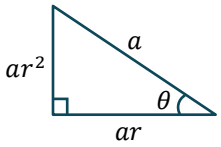
$r^2 = \dfrac{\sqrt{5}+1}{2}$; $r = \sqrt{\dfrac{\sqrt{5}+1}{2}}$

tangent of smallest angle = $\tan\theta = \dfrac{1}{r} = \sqrt{\left(\dfrac{2}{\sqrt{5}+1}\right)}$

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Case - II

$0 < r < 1$



$a^2 = a^2r^2 + a^2r^4 \Rightarrow r^4 + r^2 - 1 = 0$

$r^2 = \frac{\sqrt{5}-1}{2}$; $r = \sqrt{\frac{\sqrt{5}-1}{2}}$

tangent of smallest angle = $\tan\theta = r = \sqrt{\frac{\sqrt{5}-1}{2}}$

Illustration 16

If b_1, b_2, b_3 ($b_i > 0$) are three successive terms of a G.P. with common ratio r, the value of r for which the inequality $b_3 > 4b_2 - 3b_1$ holds is given by

(A) $r > 3$ (B) $0 < r < 1$ (C) $r = 3.5$ (D) $r = 5.2$

Ans. (A,B,C,D)

Solution:

b_1, b_2, b_3 are in G.P.

$\therefore b_3 > 4b_2 - 3b_1 \Rightarrow r^2 > 4r - 3$

$\Rightarrow r^2 - 4r + 3 > 0 \Rightarrow (r-1)(r-3) > 0$ So $0 < r < 1$ and $r > 3$

Comprehension For (Illustration No 17 to 18)

Let $\Delta^1T_n = T_{n+1} - T_n$, $\Delta^2 T_n = \Delta^1T_{n+1} - \Delta^1T_n$, $\Delta^3T_n = \Delta^2T_{n+1} - \Delta^2T_n$, ... , and so on, where $T_1, T_2, T_3, ...T_{n-1}$, T_n, T_{n+1},..... are the terms of infinite G.P. whose first term is a natural number and common ratio is equal to 'r'.

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Illustration 17

If $\Delta^2 T_1 = 36$, then sum of all possible integral values of r is equal to :

(A) 8 (B) 4 (C) 5 (D) –2

Ans. (A)

Solution:

Let first term is 'a'

$a(1-r)^2 = 36$

$\Rightarrow r$ can be 2, 3, 4, 7, –1, –2, –5

Illustration 18

Let $\sum_{n=1}^{\infty} T_n = \frac{7}{3}$ and $r = \frac{p}{7}$ then sum of squares of all possible value of p is equal to:

(A) 42 (B) 46 (C) 45 (D) 30

Ans. (B)

Solution:

$\frac{a}{1-r} = \frac{7}{3} \Rightarrow r = \frac{7-3a}{7} \Rightarrow a$ can be 1, 2, 3, 4 and p can be 4, 1, –2, –5

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Illustration 19

If $\Delta^7 T_n = \Delta^3 T_n$, then '$r$' equal to

(A) 2 (B) 4 (C) 7 (D) –2

Ans. (A)

Solution:

$ar^{n-1}(1-r)^6 = ar^{n-1}(1-r)^2 \Rightarrow (1-r)^4 = 1 \Rightarrow r = 2$

Illustration 20

The sum of the first ten terms of an AP is 155 & the sum of first two terms of a GP is 9. The first term of the AP is equal to the common ratio of the GP & the first term of the GP is equal to the common difference of the AP. Find the two progressions.

Solution:

(3 + 6 + 12 +.....) ; (2/3 + 25/3 + 625/6 +.....) G.P.

$(2 + 5 + 8 + ...) ; \left(\frac{25}{2}+\frac{79}{6}+\frac{83}{6}+.....\right)$ A.P.

Let first term of $AP = a$ and common difference = d

$\therefore \frac{10}{2}[2a + 9d] = 155 \Rightarrow .2a + 9d = 31$

GP first term = a' and common ratio = r

$a' + a'r = 9 \Rightarrow a'(1 + r) = 9$

given $a = r$ and $a' = d \Rightarrow 2r + 9a' = 31$

$\Rightarrow 2r + 9 . \frac{9}{1+r} = 31 \Rightarrow 2r^2 - 29r + 50 = 0 \Rightarrow r = 2, 25/2.$

(i) if $r = 2$, So $a = 2 \Rightarrow a' = 3 = d$

(ii) if $r = \frac{25}{2} \Rightarrow a = \frac{25}{2} \Rightarrow a' = \frac{9}{\frac{27}{2}} = \frac{2}{3} = d$

So the series 3, 6, 12..... second $\frac{2}{3}, \frac{25}{3}, \frac{625}{6}$

6. HARMONIC PROGRESSION (H.P.):

A sequence is said to be in H.P. if the reciprocal of its terms are in AP.

If the sequence $a_1, a_2, a_3, ..., a_n$ is an HP then $1/a_1, 1/a_2, ..., 1/a_n$ is an AP . Here we do not have the formula for the sum of the n terms of an HP. The general form of a harmonic progression is $\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, ... \frac{1}{a+(n-1)d}$

Note :

No term of any H.P. can be zero.

(i) If a, b, c are in HP, then $b = \frac{2ac}{a+c}$ or $\frac{a}{c} = \frac{a-b}{b-c}$

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Illustration 21

The sum of three numbers in H.P. is 37 and the sum of their reciprocals is 1/4. Find the numbers.

Solution:

Three number in H.P. can be taken as

$\frac{1}{a-d}, \frac{1}{a}, \frac{1}{a+d}$

then $\frac{1}{a-d}+\frac{1}{a}+\frac{1}{a+d} = 37$ ...(i)

and $a - d + a + a + d = \frac{1}{4} \Rightarrow a = \frac{1}{12}$

from (i), $\frac{12}{1-12d}+12+\frac{12}{1+12d}=37 \Rightarrow \frac{12}{1-12d}+\frac{12}{1+12d}=25$

$\Rightarrow \frac{24}{1-144d^2}=25$

$\Rightarrow 1-144d^2=\frac{24}{25}$

$\Rightarrow d^2 = \frac{1}{25\times144}$

$\therefore\ d=\pm\frac{1}{60}$

$\therefore\ a-d, a, a+d$ are $\frac{1}{15}, \frac{1}{12}, \frac{1}{10}$ or $\frac{1}{10}, \frac{1}{12}, \frac{1}{15}$

Hence, three numbers in H.P. are 15, 12, 10 or 10, 12, 15 **Ans.**

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Illustration 22

Suppose a is a fixed real number such that $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz}$

If p, q, r are in A.P., then prove that x, y, z are in H.P.

Solution:

$\because\ p, q, r$ are in A.P.

$\therefore\ q - p = r - q$ ...(i)

$\Rightarrow p - q = q - r = k$ (let)

given $\frac{a-x}{px}=\frac{a-y}{qy}=\frac{a-z}{rz} \Rightarrow \frac{\frac{a}{x}-1}{p}=\frac{\frac{a}{y}-1}{q}=\frac{\frac{a}{z}-1}{r}$

$\Rightarrow \frac{\left(\frac{a}{x}-1\right)-\left(\frac{a}{y}-1\right)}{p-q}=\frac{\left(\frac{a}{y}-1\right)-\left(\frac{a}{z}-1\right)}{q-r}$ (by law of proportion)

$\Rightarrow \frac{\frac{a}{x}-\frac{a}{y}}{k}=\frac{\frac{a}{y}-\frac{a}{z}}{k}$ {from (i)}

$\Rightarrow a\left(\frac{1}{x}-\frac{1}{y}\right)=a\left(\frac{1}{y}-\frac{1}{z}\right) \Rightarrow \frac{1}{x}-\frac{1}{y}=\frac{1}{y}-\frac{1}{z}$

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$\therefore \frac{2}{y}=\frac{1}{x}+\frac{1}{z}$

$\therefore \frac{1}{x},\frac{1}{y},\frac{1}{z}$ are in A.P.

Hence x, y, z are in H.P.

Illustration 23

If a_1, a_2, ... are in H.P. and $f(k) = \sum_{r=1}^{n}(a_r - a_k)$, then $\frac{a_1}{f(1)},\frac{a_2}{f(2)},.....\frac{a_n}{f(n)}$ are in

(A) A.P. (B) G.P. (C) H.P. (D) None of these

Ans. (C)

Solution:

$f(k)\ \sum_{r=1}^{n}(a_r - a_k)$

$= \sum_{r=1}^{n}a_r - \sum_{r=1}^{n}a_k$

$f(k) = \lambda - na_k$

$f(i) = \lambda - na_i$

$\frac{a_i}{f(i)}=\frac{a_i}{\lambda - na_i}$

$= \frac{1}{\frac{\lambda}{a_i}-n}$

$<a_i>$ in $A.P.$

$\Rightarrow <\frac{1}{a_i}>$ in A.P.

$\Rightarrow <\frac{\lambda}{a_i}-n>$ in A.P.

$\Rightarrow <\frac{1}{\frac{\lambda}{a_i}-n}>$ in H.P.

$\Rightarrow <\frac{a_i}{f(i)}>$ in A.P.

Illustration 24

Given that α, γ are roots of the equation $Ax^2 - 4x + 1 = 0$ and β, δ the roots of the equation $Bx^2 - 6x + 1 = 0$, then find value of $(A + B)$, such that α, β, γ & δ are in H.P.

Ans. (11)

Solution:

$\alpha + \gamma = \frac{4}{A}$, $\alpha\gamma = \frac{1}{A}$ and $\beta + \delta = \frac{6}{B}$, $\beta\delta = \frac{1}{B}$

Since $\alpha, \beta, \gamma, \delta$ are in H.P. $\beta = \frac{2\alpha\gamma}{\alpha+\gamma} = \frac{1}{2}$ is root of $Bx^2 - 6x + 1 = 0 \Rightarrow B = 8$

similarly $\gamma = \frac{2\beta\delta}{\beta+\delta} = \frac{1}{3}$ is root of $Ax^2 - 4x + 1 = 0 \Rightarrow A = 3$

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Illustration 25

Let a, x, b be in A.P; a, y, b be in G.P and a, z, b be in H.P. If $x = y + 2$ and $a = 5z$, then

(A) $y^2 = xz$ (B) $x > y > z$ (C) $a = 9, b = 1$ (D) $a = 1/4, b = 9/4$

Ans. (A,B,C)

Solution:

$2x = a + b$ ...(1)

$y^2 = ab$ ...(2)

$z = \frac{2ab}{a+b}$ ...(3)

$x = y + 2$ ...(4)

and $a = 5z$ ...(5)

$z = \frac{2y^2}{2x} \Rightarrow y^2 = xz \quad \because x = y + 2$

$\therefore \frac{a+b}{2} = \sqrt{ab} + 2$ ...(6)

and $a = 5\frac{(2ab)}{a+b} \Rightarrow (a + b) = 10b$

$\Rightarrow a = 9b$ ...(7)

$\frac{9b+b}{2} = \sqrt{b.9b} + 2 \Rightarrow 5b = 3b + 2 \Rightarrow b = 1$

So $a = 9 \Rightarrow x > y > z$

Illustration 26

The value of $x + y + z$ is 15 if a, x, y, z, b are in AP while the value of $(1/x) + (1/y) + (1/z)$ is $5/3$ if a, x, y, z, b are in HP. Find a and b.

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Ans. ($a = 1, b = 9$)

Solution:

$x + y + z = 15$ ...(i)

a, x, y, z, b are in AP

Suppose d is common difference $d = \frac{b-a}{4}$

$\therefore x = a + \frac{b-a}{4} = \frac{b+3a}{4}, y = \frac{2b+2a}{4}$ and $z = \frac{3b+a}{4}$

on substituting the values of X, Y and Z in (i), we get

$\Rightarrow \frac{6a+6b}{4} = 15$

$a + b = 10$ ...(ii)

$\because \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{5}{3}$ ...(iii)

And a, x, y, z, b are in H.P.

$\therefore \frac{1}{a}, \frac{1}{x}, \frac{1}{y}, \frac{1}{z}, \frac{1}{b}$ are in A.P.

$\therefore \frac{1}{x} = \frac{1}{a} + \frac{\left(\frac{1}{b} - \frac{1}{a}\right)}{4}, \frac{1}{y} = \frac{1}{a} + \frac{2}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$ and $\frac{1}{z} = \frac{1}{a} + \frac{3}{4}\left(\frac{1}{b} - \frac{1}{a}\right)$

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on substituting the value of $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ in (iii), we get

$\frac{3}{a}+\frac{6}{4}\left(\frac{1}{b}-\frac{1}{a}\right)=\frac{5}{3}$

$\Rightarrow \frac{3}{2a}+\frac{3}{2b}=\frac{5}{3} \Rightarrow \frac{1}{a}+\frac{1}{b}=\frac{10}{9}$ ...(iv)

By equations (ii) & (iv), we get

$a = 9, b = 1$ or $a = 1, b = 9$

7. MEANS

(a) Arithmetic Mean:

If three terms are in A.P. then the middle term is called the A.M. between the other two, so if a, b, c are in A.P., b is A.M. of a & c. So A.M. of a and $c = \frac{a+c}{2}=b$.

Insertion of n-Arithmetic Means Between Two Numbers:

If a, b be any two given numbers & $a, A_1, A_2, ..., A_n, b$ are in AP, then $A_1, A_2, ... A_n$ are the 'n' A.M's between a & b then. $A_1 = a + d$, $A_2 = a + 2d$,....., $A_n = a + nd$ or $b - d$, where $d=\frac{b-a}{n+1}$

$\Rightarrow A_1=a+\frac{b-a}{n+1}$, $A_2=a+\frac{2(b-a)}{n+1}$ .....

Note: Sum of n A.M's inserted between a & b is equal to n times the single A.M. between a & b

i.e. $\sum_{r=1}^{n} A_r = nA$ where A is the single A.M. between a & b.

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(b) Geometric Mean:

If a, b, c are in G.P., then b is the G.M. between a & c, $b^2 = ac$. So G.M. of a and $c = \sqrt{ac}=b$

Insertion of n-Geometric Means Between Two Numbers:

If a, b are two given positive numbers & $a, G_1, G_2, ..., G_n, b$ are in G.P. Then $G_1, G_2, G_3, ...G_n$ are 'n' G.Ms between a & b. where $b = ar^{n+1} \Rightarrow r = (b/a)^{1/n+1}$

$G_1 = a(b/a)^{1/n+1}$, $G_2 = a(b/a)^{2/n+1}$ ....., $G_n = a(b/a)^{n/n+1}$

$= ar$, $= ar^2$, ... $= ar^n = b/r$

Note : The product of n G.Ms between a & b is equal to n^{th} power of the single G.M. between a & b

i.e., $\prod_{r=1}^{n} G_r = (G)^n$ where G is the single G.M. between a & b

(c) Harmonic Mean:

If a, b, c are in H.P., then b is H.M. between a & c. So, H.M. of a and $c = \frac{2ac}{a+c} = b$.

Insertion of 'n' HM's Between a and b:

$a, H_1, H_2, H_3, ..., H_n, b \rightarrow$ H.P

$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, ... \frac{1}{H_n}, \frac{1}{b} \rightarrow A.P.$

$\frac{1}{b}=\frac{1}{a}+(n+1)D \quad \Rightarrow D=\frac{\frac{1}{b}-\frac{1}{a}}{n+1}$

$\frac{1}{H_n}=\frac{1}{a}+n\left(\frac{\frac{1}{b}-\frac{1}{a}}{n+1}\right)$

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Important note:

(i) If A, G, H, are respectively A.M., G.M., H.M. between two positive number a & b then
G^2 = AH (A, G, H constitute a GP)

(ii) Let $a_1, a_2, \ldots\ldots, a_n$ be n positive real numbers, then we define their arithmetic mean (A), geometric mean (G) and harmonic mean (H) as $A = \frac{a_1 + a_2 + \ldots\ldots + a_n}{n}$

$$G = (a_1\, a_2 \ldots\ldots a_n)^{1/n} \text{ and } H = \frac{n}{\left(\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \ldots\ldots \frac{1}{a_n}\right)}$$

Illustration 27

If the arithmetic mean of two numbers a & b ($0 < a < b$) is 6 and their geometric mean G and harmonic mean H satisfy the relation $G^2 + 3H = 48$. Then find the value of $(2a - b)$

Ans. (0)

Solution:

$\frac{a+b}{2} = 6$

$G^2 + 3H = 48 \Rightarrow ab + 3\,\frac{2ab}{a+b} = 48 \Rightarrow ab + \frac{3ab}{6} = 48$

$\Rightarrow \frac{3}{2}ab = 48 \Rightarrow ab = 32 \Rightarrow a = 4, b = 8.$

Illustration 28

If the arithmetic mean of two positive numbers a & b ($a > b$) is twice their geometric mean, then a: b is:

(A) $2 + \sqrt{3} : 2 - \sqrt{3}$ (B) $7 + 4\sqrt{3} : 1$ (C) $1: 7 - 4\sqrt{3}$ (D) $2: \sqrt{3}$

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Ans. (A,B,C)

Solution:

$\frac{\frac{a+b}{2}}{\sqrt{ab}} = \frac{2}{1} \Rightarrow \frac{a+b}{2\sqrt{ab}} = \frac{2}{1}$ use compendendo and dividendo rule

$\Rightarrow \frac{a+b+2\sqrt{ab}}{a+b-2\sqrt{ab}} = \frac{3}{1} \Rightarrow \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}} = \frac{\sqrt{3}}{1} \Rightarrow \frac{2\sqrt{a}}{2\sqrt{b}} = \frac{\sqrt{3}+1}{\sqrt{3}-1}$

$\Rightarrow \frac{a}{b} = \frac{3+1+2\sqrt{3}}{3+1-2\sqrt{3}} \Rightarrow \frac{a}{b} = \frac{2+\sqrt{3}}{2-\sqrt{3}} = \frac{\left(2+\sqrt{3}\right)\left(2+\sqrt{3}\right)}{4-3} = 7 + 4\sqrt{3}$ **Ans**

Illustration 29

Let a, b be positive real numbers. If a, A_1, A_2, b are in arithmetic progression, a, G_1, G_2, b are in geometric progression and a, H_1, H_2, b are in harmonic progression, show that

$\frac{G_1\, G_2}{H_1\, H_2} = \frac{A_1 + A_2}{H_1 + H_2} = \frac{(2a + b)\,(a + 2b)}{9\,ab}$.

Solution:

$a, A_1, A_2, b \rightarrow$ A.P. $\Rightarrow A_1 + A_2 = a + b.$

$a, G_1, G_2, b \rightarrow$ G.P. $\Rightarrow G_1\, G_2 = ab$

$a, H_1, H_2, b \rightarrow$ H.P. $\Rightarrow \frac{ab}{a+b} = \frac{H_1\ H_2}{H_1 + H_2}$

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$\Rightarrow \frac{A_1+A_2}{H_1+H_2}=\frac{G_1\ G_2}{H_1\ H_2} \Rightarrow \frac{1}{H_1}=\frac{1}{a}+d$

$\frac{1}{b}=\frac{1}{a}+3d \Rightarrow \frac{a-b}{3\ a\ b}=d$

$\therefore \frac{1}{H_1}=\frac{a+2b}{3ab}$ or $\frac{1}{H_2}=\frac{2a+b}{3\ ab}$

$\frac{1}{H_1H_2}=\frac{(a+2b)\ (2a+b)}{9a^2b^2} \Rightarrow \frac{G_1G_2}{H_1\ H_2}=\frac{(a+2b)\ (2a+b)}{9\ ab}$

Illustration 30

Let a_1, a_2,....., a_n be positive real numbers in geometric progression. For each n, let A_n, G_n, H_n be respectively the arithmetic mean, geometric mean & harmonic mean of a_1, a_2,....., a_n. Prove that $G=\prod_{K=1}^{n}(A_k\ \ H_k)^{\frac{1}{2n}}$, Where G is geometric mean between G_1, G_2, ..., G_n.

Solution:

Let G_m be the geometric mean of G_1, G_2,.....,$G_n \Rightarrow G_m=(G_1\cdot G_2.....G_n)^{1/n}$

$=[(a_1)(a_1a_1r)^{1/2}.(a_1.a_1r.a_1r^2)^{1/3}...\ (a_1.a_1r.a_1r^2\ ...a_1r^{n-1})^{1/n}]^{1/n}$

where r is the common ratio of G.P. a_1, a_2, ...,a_n. $=[(a_1.a_1...n \text{ times})\ (r^{1/2}.r^{3/3}.r^{6/4}....r^{\frac{(n-1)n}{2n}})]^{1/n}$

$\Rightarrow a_1.\left[r^{\frac{1}{2}\left[\frac{(n-1)\ n}{2}\right]}\right]^{1/n}=a_1\left[r^{\frac{n-1}{4}}\right]$. Now, $A_n=\frac{a_1+a_2+.....+a_n}{n}=\frac{a_1(1-r^n)}{n\ (1-r)}$

and $H_n=\frac{n}{\left(\frac{1}{a_1}+\frac{1}{a_2}...+\frac{1}{a_n}\right)}=\frac{n}{\frac{1}{a_1}\left(1+\frac{1}{r}+...+\frac{1}{r^{n-1}}\right)}=\frac{a_1\ n(1-r)\ \ r^{n-1}}{1-r^n}$

Again $A_n.H_n=\frac{a_1(1-r^n)}{n\ (1-r)}\times\frac{a_1n\ (1-r)\ r^{n-1}}{(1-r^n)}=a_1^2\ \ r^{n-1} \Rightarrow \prod_{k=1}^{n}A_kH_k=\prod_{k=1}^{n}(a_1^2\ \ r^{k-1})$

$=(a_1^2.\ a_1^2...\ n\ \text{ times})\times(r^0.r^1.r^2\ ...r^{n-1})=a_1^{2n}.r^{1+2+....(n-1)}=a_1^{2n}\ \ r^{\frac{n(n-1)}{2}}=[a_1\ \ r^{\frac{n-1}{4}}]^{2n}$

$=[G_m]^{2n} \Rightarrow G_m=\left[\prod_{k=1}^{n}A_KH_K\right]^{1/2n} \Rightarrow G_m=(A_1A_2\ ...\ A_n.H_1H_2\ ....H_n)^{1/2n}$

Illustration 31

If n is a root of the equation $x^2(1-ac)-x(a^2+c^2)-(1+ac)=0$ and if n HM's are inserted between a and c, show that the difference between the first and the last mean is equal to $ac(a-c)$.

Solution:

Let n HM's between a and c are H_1, H_2, H_3 , ... H_n

$\therefore \frac{1}{a},\frac{1}{H_1},\frac{1}{H_2},.....,\frac{1}{H_n},\frac{1}{c}$ are in A.P.

suppose 'd' is common difference of this A.P.

$\therefore \frac{1}{c}=\frac{1}{a}+(n+1)\ d \Rightarrow d=\frac{1}{(n+1)}\left(\frac{1}{c}-\frac{1}{a}\right)$

So $\frac{1}{H_1}=\frac{1}{a}+\frac{1}{(n+1)}\left(\frac{1}{c}-\frac{1}{a}\right)=\frac{nc+a}{(n+1)\ ca} \Rightarrow H_1=\frac{(n+1)\ ca}{nc+a}$

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And $\frac{1}{H_n} = \frac{1}{a} + \frac{n}{(n+1)}\left(\frac{1}{c} - \frac{1}{a}\right) = \frac{c+na}{(n+1)\ ca} \Rightarrow H_n = \frac{(n+1)\ ca}{c+na}$

$\therefore H_1 - H_n = ac(n+1)\left[\frac{1}{nc+a} - \frac{1}{c+na}\right] = ac(n+1)\left[\frac{(a-c)\ (n-1)}{(nc+a)\ (c+na)}\right]$

$H_1 - H_n = \frac{ac\ (n^2-1)\ (a-c)}{n^2ac+n(a^2+c^2)+ac}$ ...(i)

$\because$ n is a root of $x^2(1-ac) - x(a^2+c^2) - (1+ac) = 0$

we have $n^2(1-ac) - n(a^2+c^2) - (1+ac) = 0$

$\Rightarrow n^2ac + n(a^2+c^2) + ac = (n^2-1)$ ...(ii)

$\therefore$ from (i) and (ii), we get

So $H_1 - H_n = ac(a-c)$ Ans.

Illustration 32

Let A, G, H be A.M., G.M. and H.M. of three positive real numbers a, b, c respectively such that $G^2 = AH$, then prove that a, b, c are terms of a GP.

Solution:

Given $(abc)^{2/3} = \frac{(a+b+c)}{3} \cdot \frac{3}{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}} \Rightarrow (ab+bc+ac)^3 = abc(a+b+c)^3$ ...(i)

Now consider the polynomial $p(x) = x^3 + mx^2 + nx + p$, with roots a, b, c then have

$a+b+c = -m$; $ab+bc+ca = n$; $abc = -p$

using these values equation (i) becomes $n^3 = m^3p$ ...(ii)

Hence, if $m \neq 0$, then equation $p(x) = 0$ can be written as $x^3 + mx^2 + nx + \frac{n^3}{m^3} = 0$

Or $m^3x^3 + m^4x^2 + nm^3x + n^3 = 0 \Rightarrow (mx+n)(m^2x^2 + (m^3 - mn)x + n^2) = 0$

It follows that one of the roots of $p(x) = 0$ is $x_1 = -\frac{n}{m}$ and other two satisfy the condition $x_2x_3 = \frac{n^2}{m^2}$

$\Rightarrow x_1^2 = x_2\,x_3$

Thus the roots are the terms of a geometric sequence.

If should be noted that $m, n \neq 0$ as in this case $x^3 + p = 0$ cannot have three real roots.

8. ARITHMETICO - GEOMETRIC PROGRESSION:

A series, each term of which is formed by multiplying the corresponding term of an A.P. & G.P. is called the Arithmetico-Geometric Series , e.g. $1 + 3x + 5x^2 + 7x^3 + ...$

Here 1, 3, 5, ... are in A.P. & 1, x, x^2, x^3 ..... are in G.P.

(a) Sum of n Terms of an Arithmetico-Geometric Progression:

Let $S_n = a + (a+d)r + (a+2d)r^2 + ... + [a+(n-1)d]r^{n-1}$

then $S_n = \frac{a}{1-r} + \frac{dr(1-r^{n-1})}{(1-r)^2} - \frac{[a+(n-1)d]\ r^n}{1-r}$, $r \neq 1$

(b) Sum to Infinity:

If $0 < |r| < 1$ & $n \to \infty$, then $\lim_{n\to\infty} r^n = 0, S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$

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Illustration 33

Find the sum of series $4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \ldots\ldots\ldots \infty$, $|x| < 1$

Solution:

Let $S = 4 - 9x + 16x^2 - 25x^3 + 36x^4 - 49x^5 + \ldots\ldots\ldots \infty$

$-Sx = -4x + 9x^2 - 16x^3 + 25x^4 - 36x^5 + \ldots\ldots\ldots \infty$

On subtraction, we get

$S(1 + x) = 4 - 5x + 7x^2 - 9x^3 + 11x^4 - 13x^5 + \ldots\ldots\ldots \infty$

$-S(1 + x)x = -4x + 5x^2 - 7x^3 + 9x^4 - 11x^5 + \ldots\ldots\ldots \infty$

On subtraction, we get

$S(1 + x)^2 = 4 - x + 2x^2 - 2x^3 + 2x^4 - 2x^5 + \ldots\ldots\ldots \infty$

$= 4 - x + 2x^2 (1 - x + x^2 - \ldots\ldots\ldots\infty) = 4 - x + \dfrac{2x^2}{1+x} = \dfrac{4+3x+x^2}{1+x}$

$S = \dfrac{4+3x+x^2}{(1+x)^3}$ **Ans.**

Illustration 34

Find the sum of series up to n terms $\left(\dfrac{2n+1}{2n-1}\right) + 3\left(\dfrac{2n+1}{2n-1}\right)^2 + 5\left(\dfrac{2n+1}{2n-1}\right)^3 + \ldots\ldots\ldots$.

Solution:

Let $x = \left(\dfrac{2n+1}{2n-1}\right)$

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For $x \neq 1$, let

$S = x + 3x^2 + 5x^3 + \ldots\ldots\ldots + (2n - 3)x^{n-1} + (2n - 1)x^n$ ...(i)

$\Rightarrow xS = x^2 + 3x^3 + \ldots\ldots\ldots + (2n - 5)x^{n-1} + (2n - 3)x^n + (2n - 1)x^{n+1}$ ...(ii)

Subtracting (ii) from (i), we get

$(1 - x)S = x + 2x^2 + 2x^3 + \ldots\ldots\ldots + 2x^{n-1} + 2x^n - (2n - 1)x^{n+1}$

$= x + \dfrac{2x^2(1-x^{n-1})}{1-x} - (2n-1)x^{n+1}$

$= \dfrac{x}{1-x}[1 - x + 2x - 2x^n - (2n - 1)x^n + (2n - 1)x^{n+1}]$

$\Rightarrow S = \dfrac{x}{(1-x)^2}[(2n-1)x^{n+1} - (2n + 1)x^n + 1 + x]$

Thus $\left(\dfrac{2n+1}{2n-1}\right) + 3\left(\dfrac{2n+1}{2n-1}\right)^2 + \ldots\ldots\ldots + (2n-1)\left(\dfrac{2n+1}{2n-1}\right)^n$

$= \left(\dfrac{2n+1}{2n-1}\right)\left(\dfrac{2n-1}{2}\right)^2\left[(2n-1)\left(\dfrac{2n+1}{2n-1}\right)^{n+1} - (2n+1)\left(\dfrac{2n+1}{2n-1}\right)^n + 1 + \dfrac{2n+1}{2n-1}\right]$

$= \dfrac{4n^2-1}{4} \cdot \dfrac{4n}{2n-1} = n(2n + 1)$ **Ans.**

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9. SIGMA NOTATIONS (Σ):

THEOREMS:

(a) $\sum_{r=1}^{n}(a_r \pm b_r)=\sum_{r=1}^{n}a_r \pm \sum_{r=1}^{n}b_r$ **(b)** $\sum_{r=1}^{n}k\,a_r = k\sum_{r=1}^{n}a_r$ **(c)** $\sum_{r=1}^{n}k = nk$; where k is a constant.

10. RESULTS:

(a) $\sum_{r=1}^{n}r = \frac{n(n+1)}{2}$ (sum of the first n natural numbers)

(b) $\sum_{r=1}^{n}r^2 = \frac{n(n+1)(2n+1)}{6}$ (sum of the squares of the first n natural numbers)

(c) $\sum_{r=1}^{n}r^3 = \frac{n^2(n+1)^2}{4} = \left[\sum_{r=1}^{n} r\right]^2$ (sum of the cubes of the first n natural numbers)

(d) $\sum_{r=1}^{n}r^4 = \frac{n}{30}(n+1)(2n+1)(3n^2+3n-1)$

(e) $\sum_{r=1}^{n}(2r-1)=n^2$ (sum of first n odd natural numbers)

(f) $\sum_{r=1}^{n}2r = n(n+1)$ (sum of first n even natural numbers)

Note:

If n^{th} term of a sequence is given by $T_n = an^3 + bn^2 + cn + d$ where a, b, c, d are constants, then sum of n terms $S_n = \Sigma T_n = a\Sigma n^3 + b\Sigma n^2 + c\Sigma n + \Sigma d$

This can be evaluated using the above results.

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Illustration 35

Find the sum of the series to n terms whose n^{th} terms is $3n + 2$.

Solution:

$\Rightarrow S_n = \Sigma T_n = \Sigma(3n + 2) = 3\Sigma n + \Sigma 2 = \frac{3(n+1)n}{2}+2n=\frac{n}{2}(3n+7)$

Illustration 36

If $T_k = k^3 + 3^k$, then find $\sum_{k=1}^{n}T_k$.

Solution:

$\Rightarrow \sum_{k=1}^{n}T_k = \sum_{k=1}^{n}k^3+\sum_{k=1}^{n}3^k = \left(\frac{n(n+1)}{2}\right)^2+\frac{3(3^n-1)}{3-1}=\left(\frac{n(n+1)}{2}\right)^2+\frac{3}{2}(3^n-1)$

Illustration 37

Find the value of the expression $\sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1$

Solution:

$\Rightarrow \sum_{i=1}^{n}\sum_{j=1}^{i}\sum_{k=1}^{j}1 = \sum_{i=1}^{n}\sum_{j=1}^{i}j = \sum_{i=1}^{n}\frac{i(i+1)}{2}=\frac{1}{2}\left[\sum_{i=1}^{n}i^2+\sum_{i=1}^{n}i\right]=\frac{1}{2}\left[\frac{n(n+1)(2n+1)}{6}+\frac{n(n+1)}{2}\right]$

$=\frac{n(n+1)}{12}[2n+1+3]=\frac{n(n+1)(n+2)}{6}$.

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Illustration 38

Sum up to 16 terms of the series $\frac{1^3}{1}+\frac{1^3+2^3}{1+3}+\frac{1^3+2^3+3^3}{1+3+5}+.....$ is

(A) 450 (B) 456 (C) 446 (D) none of these

Ans. (C)

Solution:

$$t_n=\frac{1^3+2^3+3^3+....+n^3}{1+3+5+....(2n-1)}=\frac{\left\{\frac{n(n+1)}{2}\right\}^2}{\frac{n}{2}\{2+2(n-1)\}}=\frac{\frac{n^2(n+1)^2}{4}}{n^2}=\frac{(n+1)^2}{4}==\frac{n^2}{4}+\frac{n}{2}+\frac{1}{4}$$

$$\therefore\ S_n=\Sigma t_n=\frac{1}{4}\Sigma n^2+\frac{1}{2}\Sigma n+\frac{1}{4}\Sigma 1\ =\frac{1}{4}.\frac{n(n+1)(2n+1)}{6}+\frac{1}{2}.\frac{n(n+1)}{2}+\frac{1}{4}.n$$

$$\therefore\ S_{16}=\frac{16.17.33}{24}+\frac{16.17}{4}+\frac{16}{4}=446$$

Illustration 39

Let $a_1, a_2, ...a_n$, be real numbers such that

$\sqrt{a_1}+\sqrt{a_2-1}+\sqrt{a_3-2}+.....+\sqrt{a_n-(n-1)}=\frac{1}{2}\ (a_1+a_2+.....+a_n)-\frac{n(n-3)}{4}$ then find the value of $\sum_{i=1}^{100}a_i$

Ans. (5050)

Solution:

Let $\sqrt{a_1}=b_1\ ;\ \sqrt{a_2-1}=b_2\ ;\ \sqrt{a_3-2}=b_3\ ;\ .....\ \sqrt{a_n-(n-1)}=b_n$

$$\therefore\ b_1+b_2+.....+b_n=\frac{1}{2}\ [b_1^2+(b_2^2+1)+(b_3^2+2)+.....+(b_n^2+(n-1))]-\frac{n(n-3)}{4}$$

$$\therefore\ \Sigma b_1=\frac{1}{2}\ [(b_1^2+b_2^2+b_3^2+.....+b_n^2)+(1+2+3+.....+(n-1)]-\frac{n(n-3)}{4}$$

$$\Rightarrow 2\Sigma b_1=\Sigma b_1^2+\frac{n(n-1)}{2}-\frac{n(n-3)}{2}\ \Rightarrow 2\Sigma b_1=\Sigma b_1^2+n$$

$$\therefore\ \Sigma b_1^2-2\Sigma b_1+\Sigma 1=0\Rightarrow\sum_{i=1}^{n}(b_1-1)^2=0$$

$b_1-1=0\Rightarrow b_1^2=a_1=1$

$b_2-1=0\Rightarrow b_2^2=a_2-1=1\Rightarrow a_2=2$

$b_3-1=0\Rightarrow b_3^2=a_3-2=1\Rightarrow a_2=3$ and soon

hence $a_n=n$

$$\therefore\ \sum_{i=1}^{100}a_1\ =1+2+3+....+100=5050$$

Illustration 40

Find sum of the series $\frac{n}{1\,.\,2\,.\,3}+\frac{n-1}{2\,.\,3\,.\,4}+\frac{n-2}{3\,.\,4\,.\,5}+....$ up to n terms..

Ans. $\frac{n(n+1)}{4(n+2)}$

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Solution:

$\frac{n}{1.2.3}+\frac{n-1}{2.3.4}+.....+\frac{1}{n(n+1)(n+2)}$; $S_n=\sum_{r=1}^{n}\frac{n+1}{r(r+1)(r+2)}-\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}$

Now $\sum_{r=1}^{n}\frac{1}{r\ (r+1)(r+2)}=\sum_{r=1}^{n}\frac{1}{2\ (r+1)}\left[\frac{1}{r}-\frac{1}{r+2}\right]=\frac{1}{2}\left[\sum_{r=1}^{n}\frac{1}{r(r+1)}-\sum_{r=1}^{n}\frac{1}{(r+1)(r+2)}\right]$

$=\frac{1}{2}\left[\sum_{r=1}^{n}\left(\frac{1}{r}-\frac{1}{r+1}\right)-\sum_{r=1}^{n}\left(\frac{1}{r+1}-\frac{1}{r+2}\right)\right]=\frac{1}{2}\left[\left(1-\frac{1}{n+1}\right)-\left(\frac{1}{2}-\frac{1}{n+2}\right)\right]$

$=\frac{1}{2}\left[\frac{n}{n+1}-\frac{n}{2(n+2)}\right]=\frac{n}{2}\left[\frac{1}{n+1}-\frac{1}{2(n+2)}\right]$

Now $S_n=(n+1)\ \frac{n}{2}\left[\frac{1}{n+1}-\frac{1}{2(n+2)}\right]-\left(\frac{1}{2}-\frac{1}{n+2}\right)$

$=\frac{n}{2}-\frac{n(n+1)}{4(n+2)}-\frac{n}{2(n+2)}=\frac{2n^2+4n-n^2-n-2n}{4(n+2)}=\frac{n^2+n}{4(n+2)}=\frac{n(n+1)}{4(n+2)}$

11. METHOD OF DIFFERENCE:

Sometimes the nth term of a sequence or a series cannot be determined by the method, we have discussed earlier. So, we compute the difference between the successive terms of given sequence to obtain the nth term.

If $T_1, T_2, T_3,.....,T_n$ are the terms of a sequence then sometimes the terms $T_2 - T_1, T_3 - T_2,.....$ constitute an AP/GP. n^{th} term of the series is determined & the sum to n terms of the sequence can easily be obtained.

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Case 1:

(a) If difference series are in A.P., then

Let $T_n = an^2 + bn + c$, where a, b, c are constant

(b) If difference of difference series is in A.P., then

Let $T_n = an^3 + bn^2 + cn + d$, where a, b, c, d are constant

Case 2:

(a) If difference is in G.P., then

Let $T_n = ar^n + b$, where r is common ratio & a, b are constant

(b) If difference of difference is in G.P., then

Let $T_n = ar^n + bn + c$, where r is common ratio & a, b, c are constant

Determine constant by putting n = 1, 2, 3 ..... n and putting the value of T_1, T_2, T_3 ..... and sum of series $(S_n) = \sum T_n$

Illustration 41

Find the sum of n terms of the series 3 + 7 + 14 + 24 + 37 + ...

Solution:

Clearly here the differences between the successive terms are

7 – 3, 14 – 7, 24 – 14, ... i.e. 4, 7, 10, 13,....., which are in A.P.

Let $S = 3 + 7 + 14 + 24 + ... + T_n$

$S = 3 + 7 + 14 + ... + T_{n-1} + T_n$

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Subtracting, we get

$0 = 3 + [4 + 7 + 10 + 13 + \ldots\ldots (n-1) \text{ terms}] - T_n$

$\therefore\ T_n = 3 + S_{n-1}$ of an A.P. whose $a = 4$ and $d = 3$.

$$\therefore\ T_n = 3 + \left(\frac{n-1}{2}\right)(2.4+(n-2)3) = \frac{6+(n-1)(3n+2)}{2} \text{ or, } T_n = \frac{1}{2}(3n^2 - n + 4)$$

Now putting $n = 1, 2, 3, \ldots\ldots, n$ and adding

$$\therefore\ S_n = \frac{1}{2}[3\Sigma n^2 - \Sigma n + 4n] = \frac{1}{2}\left[3\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n\right] = \frac{n}{2}(n^2 + n + 4) \textbf{ Ans.}$$

Aliter Method:

Let $T_n = an^2 + bn + c$

Now, $T_1 = 3 = a + b + c$ ...(i)

$T_2 = 7 = 4a + 2b + c$ ...(ii)

$T_3 = 14 = 8a + 3b + c$ ...(iii)

Solving (i), (ii) & (iii) we get

$$a = \frac{3}{2}, b = -\frac{1}{2} \text{ \& } c = 2 \qquad \therefore T_n = \frac{1}{2}(3n^2 - n + 4)$$

$$\Rightarrow S_n = \Sigma T_n = \frac{1}{2}[3\Sigma n^2 - \Sigma n + 4n] = \frac{1}{2}\left[3\frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 4n\right] = \frac{n}{2}(n^2 + n + 4) \textbf{ Ans.}$$

Illustration 42

Find the sum of n-terms of the series $1 + 4 + 10 + 22 + \ldots\ldots$

Solution:

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Let $S = 1 + 4 + 10 + 22 + \ldots\ldots + T_n$ ...(i)

$S = 1 + 4 + 10 + \ldots\ldots + T_{n-1} + T_n$ ...(ii)

(i) – (ii) $\Rightarrow T_n = 1 + (3 + 6 + 12 + \ldots\ldots + T_n - T_{n-1})$

$$T_n = 1 + 3\left(\frac{2^{n-1} - 1}{2-1}\right)$$

$T_n = 3 \cdot 2^{n-1} - 2$

So $S_n = \Sigma T_n = 3\Sigma 2^{n-1} - \Sigma 2$

$= 3\left(\frac{2^n - 1}{2-1}\right) - 2n = 3.2^n - 2n - 3$ **Ans.**

Aliter Method:

Let $T_n = ar^n + b$, where $r = 2$

Now $T_1 = 1 = ar + b$ ...(i)

$T_2 = 4 = ar^2 + b$ ...(ii)

Solving (i) & (ii), we get

$$a = \frac{3}{2}, b = -2$$

$\therefore\ T_n = 3.2^{n-1} - 2$

$\Rightarrow S_n = \Sigma T_n = 3\Sigma 2^{n-1} - \Sigma 2$

$= 3\left(\frac{2^n - 1}{2-1}\right) - 2n = 3.2^n - 2n - 3$ **Ans.**

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Illustration 43

Find the general term and sum of n terms of the series $1 + 5 + 19 + 49 + 101 + 181 + 295 + ...$

Solution:

The sequence of difference between successive term 4, 14, 30, 52, 80 .....

The sequence of the second order difference is 10, 16, 22, 28, ... clearly it is an A.P.

So, let n^{th} term

$\Rightarrow T_n = an^3 + bn^2 + cn + d$

$\Rightarrow a + b + c + d = 1$ ...(i)

$\Rightarrow 8a + 4b + 2c + d = 5$ ...(ii)

$\Rightarrow 27a + 9b + 3c + d = 19$ ...(iii)

$\Rightarrow 64a + 16b + 4c + d = 49$ ...(iv)

from (i), (ii), (iii) & (iv)

$\Rightarrow a = 1, b = -1, c = 0, d = 1 \Rightarrow T_n = n^3 - n^2 + 1$

$$\therefore S_n = \sum(n^3 - n^2 + 1) = \left(\frac{n(n+1)}{2}\right)^2 - \frac{n(n+1)(2n+1)}{6} + n = \frac{n(n^2-1)(3n+2)}{12} + n$$

Illustration 44

If $\sum_{r=1}^{n} T_r = \frac{n}{8}(n+1)(n+2)(n+3)$, then find $\sum_{r=1}^{n} \frac{1}{T_r}$.

Solution:

$\because T_n = S_n - S_{n-1}$

$$= \sum_{r=1}^{n} T_r - \sum_{r=1}^{n-1} T_r = \frac{n(n+1)(n+2)(n+3)}{8} - \frac{(n-1)n(n+1)(n+2)}{8} = \frac{n(n+1)(n+2)}{8}[(n+3)-(n-1)]$$

$$T_n = \frac{n(n+1)(n+2)}{8}(4) = \frac{n(n+1)(n+2)}{2}$$

$$\Rightarrow \frac{1}{T_n} = \frac{2}{n(n+1)(n+2)} = \frac{(n+2)-n}{n(n+1)(n+2)} = \frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)} \quad ...(i)$$

Let $V_n = \frac{1}{n(n+1)}$

$$\therefore \frac{1}{T_n} = V_n - V_{n+1}$$

Putting $n = 1, 2, 3, ... n$

$$\Rightarrow \frac{1}{T_1} + \frac{1}{T_2} + \frac{1}{T_3} + ... + \frac{1}{T_n} = (V_1 - V_{n+1}) \Rightarrow \sum_{r=1}^{n} \frac{1}{T_r} = \frac{n^2+3n}{2(n+1)(n+2)}$$

Illustration 45

Find the sum of n terms of the series $1 . 3 . 5 + 3 . 5 . 7 + 5 . 7 . 9 + ...$

Solution:

The nth term is $(2n - 1)(2n + 1)(2n + 3)$

$T_n = (2n - 1)(2n + 1)(2n + 3)$

$$T_n = \frac{1}{8}(2n - 1)(2n + 1)(2n + 3)\{(2n + 5) - (2n - 3)\}$$

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$= \frac{1}{8}(V_n - V_{n-1})$ [Let $V_n = (2n-1)(2n+1)(2n+3)(2n+5)$]

$S_n = \sum T_n = \frac{1}{8}[V_n - V_0]$

$\therefore S_n = \frac{(2n-1)(2n+1)(2n+3)(2n+5)}{8} + \frac{15}{8} = n(2n^3 + 8n^2 + 7n - 2)$ **Ans.**

Illustration 46

Find the natural number 'a' for which $\sum_{k=1}^{n} f(a+k) = 16(2^n - 1)$, where the function f satisfied $f(x+y) = f(x).f(y)$ for all natural number x, y and further $f(1) = 2$.

Solution:

It is given that

$f(x+y) = f(x)\,f(y)$ and $f(1) = 2$

$f(1+1) = f(1)\,f(1) \Rightarrow f(2) = 2^2, f(1+2) = f(1)\,f(2) \Rightarrow f(3) = 2^3, f(2+2) = f(2)\,f(2) \Rightarrow f(4) = 2^4$

Similarly $f(k) = 2^k$ and $f(a) = 2^a$

Hence , $= \sum_{k=1}^{n} f(a+k) = \sum_{k=1}^{n} f(a) f(k) = f(a) \sum_{k=1}^{n} f(k) = 2^a \sum_{k=1}^{n} 2^k \; 2^a = (2^1 + 2^2 + \ldots\ldots + 2^n)$

$= 2^a \left\{ \frac{2(2^n - 1)}{2-1} \right\} = 2^{a+1}(2^n - 1)$

But $\sum_{k=1}^{n} f(a+k) = 16(2^n - 1)$

$2^{a+1}(2^n - 1) = 16(2^n - 1)$

$\therefore 2^{a+1} = 2^4$

$\therefore a + 1 = 4 \Rightarrow a = 3$ **Ans.**

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Illustration 47

If $S = \frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \ldots\ldots\infty$, then find the value of $14S$

Solution:

Ans. (7)

$T_n = \frac{n}{1+n^2+n^4} = \frac{1}{2}\left[\frac{(2n)}{(1+n+n^2)(1-n+n^2)}\right]$; $T_n = \frac{1}{2}\left[\frac{1}{1-n+n^2} - \frac{1}{1+n+n^2}\right]$

$T_1 = \frac{1}{2}\left[\frac{1}{1} - \frac{1}{3}\right]$; $T_2 = \frac{1}{2}\left[\frac{1}{3} - \frac{1}{7}\right]$,

$T_3 = \frac{1}{2}\left[\frac{1}{7} - \frac{1}{13}\right]$,

$\vdots$

$T_n = \frac{1}{2}\left[\frac{1}{1-n+n^2} - \frac{1}{1+n+n^2}\right]$

$S_n = \sum T_n = \frac{1}{2}\left[1 - \frac{1}{1+n+n^2}\right] = \frac{n+n^2}{2(1+n+n^2)}$

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Illustration 48

Suppose a, b, c are in A.P. and a^2, b^2, c^2 are in G.P. if $a < b < c$ and $a + b + c = \frac{3}{2}$, then the value of a is

(A) $\frac{1}{2\sqrt{2}}$ (B) $\frac{1}{2\sqrt{3}}$ (C) $\frac{1}{2} - \frac{1}{\sqrt{3}}$ (D) $\frac{1}{2} - \frac{1}{\sqrt{2}}$

Ans. (D)

Solution:

$2b = a + c$ and $b^2 = \pm ac$

case-I

if $b^2 = ac$ and $a + c + b = \frac{3}{2} \Rightarrow b = \frac{1}{2}$

$a + c = 1 \Rightarrow ac = \frac{1}{4} \Rightarrow (1 - c)\, c = \frac{1}{4}$

$c^2 - c + \frac{1}{4} = 0 \Rightarrow c = \frac{1}{2} \Rightarrow a = \frac{1}{2}$

$a = b = c$ so not valid

case-II

$b^2 = -ac$ and $b = \frac{1}{2}$; $a + c = 1 \Rightarrow ac = -\frac{1}{4}$

$(1 - c)c = -\frac{1}{4} \Rightarrow c^2 - c - \frac{1}{4} = 0 \Rightarrow c = \frac{1 \pm \sqrt{1+1}}{2} = \frac{1 \pm \sqrt{2}}{2}$

$c = \frac{1+\sqrt{2}}{2} \Rightarrow a = \frac{1-\sqrt{2}}{2}$

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Illustration 49

Consider the sequence 2, 3, 5, 6, 7, 8, 10, 11, ... of all positive integer, then 2011th term of this sequence is

(A) 2056 (B) 2011 (C) 2013 (D) 2060

Ans. (A)

Solution:

$45^2 = 2025$ & $46^2 = 2116 \Rightarrow$ there are 45 squares ≤ 2056

which are left only from sequence of possible integers since $2056 = 2011 + 45$

$\therefore$ 2011th term = 2056

Illustration 50

Let T_r and S_r be the rth term and sum up to r^{th} term of a series respectively. If for an odd number n, $S_n = n$ and $T_n = \frac{T_{n-1}}{n^2}$ then T_m (m being even) is

(A) $\frac{2}{1+m^2}$ (B) $\frac{2m^2}{1+m^2}$ (C) $\frac{(m+1)^2}{2+(m+1)^2}$ (D) $\frac{2(m+1)^2}{1+(m+1)^2}$

Ans. (D)

Solution:

$S_n - S_{n-2} = 2 \Rightarrow T_n + T_{n-1} = 2$

Also $T_n - T_{n-1} = 2$; $T_n + T_{n-1} = \left(\frac{1}{n^2} + 1\right) T_{n-1} = 2 \Rightarrow T_{n-1} = \frac{2}{1+\frac{1}{n^2}} = \frac{2n^2}{1+n^2}$

So, $T_m = \frac{2(m+1)^2}{1+(m+1)^2}$

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Illustration 51

If $1^2 + 2^2 + 3^2 + ... + 2003^2 = (2003)(4007)(334)$ and

$(1)(2003) + (2)(2002) + (3)(2001) + ... + (2003)(1) = (2003)(334)(x)$., then x equals

(A) 2005 (B) 2004 (C) 2003 (D) 2001

Ans. (A)

Solution:

If $1^2 + 2^2 + 3^2 + ... + 2003^2 = (2003)(4007)(334)$

$(1)(2003) + (2)(2002) + (3)(2001) + ... + (2003)(1) = (2003)(334)(x)$

$\Rightarrow \sum_{r=1}^{2003} r\ (2003-r+1) = (2003)(334)(x) \Rightarrow 2004.\sum_{r=1}^{2003} r - \sum_{r=1}^{2003} r^2 = (2003)(334)(x)$

$\Rightarrow 2004\left(\frac{2003\ .\ 2004}{2}\right). - 2003\ .\ (4007)\ .\ 334 = (2003)(334)(x)$

$\Rightarrow x = 2005$ **Ans.**

Illustration 52

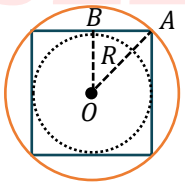
In a circle of radius R a square is inscribed, then a circle is inscribed in the square, a new square in the circle and so on for n times. If the ratio of the limit of the sum of areas of all the circles to the limit of the sum of areas of all the squares as $n \to \infty$ is k, then find the value of $\frac{4k}{\pi}$.

Ans. (2)

Solution:

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Area $= A_1 = \pi R^2 \Rightarrow OB = \frac{R}{\sqrt{2}}$



So Area $A_2 = \pi\left(\frac{R^2}{2}\right)$.

So $\lim_{n\to\infty}\left((\pi R^2) + \frac{\pi R^2}{2} + \frac{\pi R^2}{4} + ...\infty\right) = \pi R^2 \frac{1}{\left(1-\frac{1}{2}\right)}. = 2\pi R^2$

Now sum of areas of the squares $= 2R^2 + \frac{2R^2}{2} + \frac{2R^2}{4} + ... \infty = \frac{2R^2}{1-\frac{1}{2}} = 4R^2$

Illustration 53

$a, a_1, a_2, a_3, ..., a_{2n}$, b are in A.P. and $a, g_1, g_2, g_3, ...g_{2n}, b$ are in G.P. and h is the harmonic mean of a and b, if $\frac{a_1+a_{2n}}{g_1 g_{2n}} + \frac{a_2+a_{2n-1}}{g_2 g_{2n-1}} + ... \frac{a_n+a_{n+1}}{g_n g_{n+1}} +$ is equal $\frac{Kn}{20h}$ to , then find value of K.

Ans. (40)

Solution:

$a, a_1, a_2 ....., a_{2n}, b$ are in AP and $a, g_1, g_2, ...g_{2n}, b$ are in GP and $h = \frac{2ab}{a+b}$

$\therefore \frac{a_1+a_{2n}}{g_1\ \ g_{2n}} + \frac{a_2+a_{2n-1}}{g_2\ \ g_{2n-1}} + ... + \frac{a_n+a_{n+1}}{g_n\ \ g_{n+1}} = \frac{a+b}{ab} + \frac{a+b}{ab} + ... + \frac{a+b}{ab} = 2n\left(\frac{a+b}{2ab}\right).= \frac{2n}{h}$

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Illustration 54

If $\frac{25}{k} = 1^2 - \frac{2^2}{5} + \frac{3^2}{5^2} - \frac{4^2}{5^3} + \frac{5^2}{5^4} - \frac{6^2}{5^5} + \ldots\ldots \infty$, then find the value of k

Solution:

Ans. (54)

$S = 1^2 - \frac{2^2}{5} + \frac{3^2}{5^2} - \frac{4^2}{5^3} + \frac{5^2}{5^4} - \frac{6^2}{5^5} + \ldots\ldots\ldots\infty$ ...(i)

$-\frac{1}{5}S = \frac{1}{5} - \frac{2^2}{5^2} + \frac{3^2}{5^3} - \frac{4^2}{5^4} + \frac{5^2}{5^5} - + \ldots\ldots\ldots\infty$ ...(ii)

(i) – (ii) we get

$\frac{6}{5}S = 1 - \frac{3}{5} + \frac{5}{5^2} - \frac{7}{5^3} + \frac{9}{5^4} - \frac{11}{5^5} + \ldots\ldots\ldots\ldots\infty$ ...(iii)

$-\frac{6}{25}S = -\frac{1}{5} + \frac{3}{5^2} - \frac{5}{5^3} + \frac{7}{5^4} - \frac{9}{5^5} + \ldots\ldots\ldots\ldots\infty$ ...(iv)

(iii) – (iv) we get

$\frac{6s}{5}\left(\frac{6}{5}\right) = 1 - \frac{2}{5} + \frac{2}{5^2} - \frac{2}{5^3} + \frac{2}{5^4} - \ldots\ldots\ldots\infty$

$\frac{36}{25}S = 1 - \frac{2}{5}\left[\frac{1}{1+\frac{1}{5}}\right]$; $\frac{36}{25}S = 1 - \frac{2}{5}\left(\frac{5}{6}\right) = \frac{2}{3}$; $S = \frac{25}{36} \times \frac{2}{3} = \frac{25}{54}$ **Ans**

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Illustration 55

Which of the following is/are TRUE

(A) Equal numbers are always in A.P., G.P. and H.P.

(B) If a, b, c be in H.P., then $a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ will be in AP

(C) If G_1 and G_2 are two geometric means and A is the arithmetic mean inserted between two positive numbers, then the value of $\frac{G_1^2}{G_2} + \frac{G_2^2}{G_1}$ is $2A$.

(D) Let general term of a G.P. (with positive terms) with common ratio r be T_{k+1} and general term of another G.P. (with positive terms) with common ratio r be T'_{k+1}, then the series whose general term $T''_{k+1} = T_{k+1} + T'_{k+1}$ is also a G.P. with common ratio r.

Ans. (C,D)

Solution:

(A) $\because$ equal numbers are not always in A.P., G.P. and in H.P.

for example 0, 0, 0, ...

(B) $\frac{a-b}{b-c} = \frac{a}{c} \Rightarrow ac - bc = ba - ac \Rightarrow 2ac - bc = ab \Rightarrow b = \frac{2ac}{a+c}$

consider

$a - \frac{b}{2}, \frac{b}{2}, c - \frac{b}{2}$ in A.P. $\Rightarrow b - a = c - b \Rightarrow 2b = a + c$

So statement is false.

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(C) Let numbers are a, b

a, G_1, G_2, b or $A = \frac{a+b}{2}$

$r = \left(\frac{b}{a}\right)^{\frac{1}{3}}$; $G_1 = a\left(\frac{b}{a}\right)^{\frac{1}{3}}$, $G_2 = a.\left(\frac{b}{a}\right)^{\frac{2}{3}}$

$\therefore \frac{G_1^3+G_2^3}{G_1G_2} = \frac{a^3.\frac{b}{a}+a^3.\frac{b^2}{a^2}}{a^2.\frac{b}{a}} = \frac{a^2b+ab^2}{ab} = a + b = 2A$

(D) Let $T_{k+1} = ar^k$ and $T'_{k+1} = br^k$. Since $T''_{k+1} = ar^k + br^k = (a + b)r^k$,

$\therefore T''_{k+1}$ is general term of a G.P.

Illustration 56

Let $a_1, a_2, a_3, \ldots\ldots a_n$ is the sequence whose sum of first 'n' terms is represented by $S_n = an^3 + bn^2 + cn, n \in N$. If $a = \frac{a_1+a_3-xa_2}{y}$ then

(A) H.C.F of (x, y) is 2 (B) H.C.F. of (x, y) is 3

(C) L.C.M of (x, y) is 6 (D) $x + y = 8$

Ans. (A,C,D)

Solution:

$6a = (a_3 - a_2) - (a_2 - a_1) \Rightarrow a = \frac{a_1+a_3-2a_2}{6}$

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Comprehension (Illustration Q. no. 57 & 58)

In a sequence of $(4n + 1)$ terms the first $(2n + 1)$ terms are in AP whose common difference is 2, and the last $(2n + 1)$ terms are in GP whose common ratio 0.5. If the middle terms of the AP and GP are equal, then

Illustration 57

Middle term of the sequence is

(A) $\frac{n \cdot 2^{n+1}}{2^n-1}$ (B) $\frac{n \cdot 2^{n+1}}{2^{2n}-1}$ (C) $n \cdot 2^n$ (D) None of these

Ans. (A)

Illustration 58

First term of the sequence is

(A) $\frac{4n+2n \cdot 2^n}{2^n-1}$ (B) $\frac{4n-2n \cdot 2^n}{2^n-1}$ (C) $\frac{2n-n \cdot 2^n}{2^n-1}$ (D) $\frac{2n+n \cdot 2^n}{2^n-1}$

Ans. (B)

Solution: (57, 58)

Let 1st term be a. and common difference is 2 ; $T_{2n+1} = a + 4n = A$(say) $r = \frac{1}{2}$

Middle term of AP = T_{n+1} ; Middle term of GP = T_{3n+1}

$T_{n+1} = a + 2n \Rightarrow T_{3n+1} = A \cdot r^n = \frac{(a+4n)}{2^n}(a + 2n) = \frac{a+4n}{2^n} \Rightarrow 2^n a + 2n2^n = a + 4n$

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$a = \frac{4n-2n\ .\ 2^n}{2^n-1} \Rightarrow T_{2n+1} = a + 4n = \frac{4n-2n\ .\ 2^n}{2^n-1} + 4n = \frac{2n\ .\ 2^n}{2^n-1} = \frac{2^{n+1}n}{2^n-1}$

$T_{3n+1} = \frac{a+4n}{2^n} = \frac{2^{n+1}n}{2^n(2^n-1)} = \frac{2n}{2^n-1}$

Illustration 59

If total number of runs scored in n matches is $\left(\frac{n+1}{4}\right)(2^{n+1} - n - 2)$ where $n > 1$ and the runs scored in the kth match are given by $k.\ 2^{n+1-k}$, where $1 \le k \le n$, find n

Solution:

Ans. (7)

$T_k = k.2^{n+1-k}$ and $S_n = \left(\frac{n+1}{4}\right)\left(2^{n+1} - n - 2\right)$

Now, $S_n = \sum_{k=1}^{n} k.2^{n+1-k} = 2^{n+1}\sum_{k=1}^{n} k.2^{-k}$; $S_n = 2^{n+1}.\ 2\ .\left[1 - \frac{1}{2^n} - \frac{n}{2^{n+1}}\right]$ (sum of A.G.P.)

$\left(\frac{n+1}{4}\right)\left(2^{n+1} - n - 2\right) = 2.\ [2^{n+1} - 2 - n] \Rightarrow \frac{n+1}{4} = 2 \Rightarrow n = 7$

Illustration 60

Circles are inscribed in the acute angle α so that every neighbouring circles touch each other. If the radius of the first circle is R, then find the sum of the radii of the first n circles in terms of R and α.

Ans. $\frac{R\left(1 - \sin\frac{\alpha}{2}\right)}{2\sin\frac{\alpha}{2}}\left[\left(\frac{1 + \sin\frac{\alpha}{2}}{1 - \sin\frac{\alpha}{2}}\right)^n - 1\right]$

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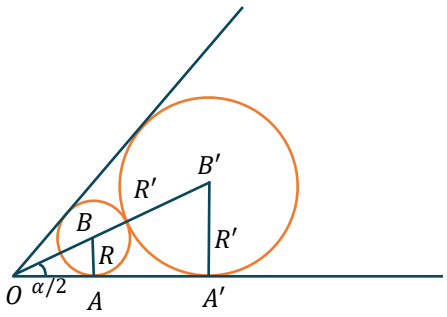
Solution:

Radius of first circle = R

Let Radius of second circle be = R'

$\frac{OB}{BA} = \text{cosec}\ \frac{\alpha}{2}$

$\therefore\ OB = R\ \text{cosec}\ \frac{\alpha}{2}$



Now $\frac{OB'}{A'\ B'} = \text{cosec}\ \frac{\alpha}{2} \Rightarrow OB' = R'\ \text{cosec}\ \frac{\alpha}{2} = OB + R + R'$

$\Rightarrow R + R\ \text{cosec}\ \frac{\alpha}{2} + R' = R'\ \text{cosec}\ \frac{\alpha}{2}$

$\Rightarrow R(1 + \text{cosec}\ \alpha/2) = R'\ (\text{cosec}\ \alpha/2 - 1)$

$\Rightarrow R' = R\left(\frac{1+\sin\alpha/2}{1-\sin\alpha/2}\right)$

if radius of the third circle be R'', then similarly $R'' = R\left(\frac{1+\sin\alpha/2}{1-\sin\alpha/2}\right)^2$

So $R + R' + R'' + ...n$ terms $= R\left[\frac{\left(\frac{1+\sin\alpha/2}{1-\sin\alpha/2}\right)^n - 1}{\left(\frac{1+\sin\alpha/2}{1-\sin\alpha/2}\right) - 1}\right] = R\left(\frac{1-\sin\alpha/2}{2\sin\alpha/2}\right)\left[\left(\frac{1+\sin\alpha/2}{1-\sin\alpha/2}\right)^n - 1\right]$

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Illustration 61

If sum of first n terms of an A.P. (having positive terms) is given by $S_n = (1 + 2T_n)(1 - T_n)$ where T_n is the nth term of series, then $T_2^2 = \frac{\sqrt{a}-\sqrt{b}}{4}$, $(a \in N, b \in N)$, then find the value of $(a + b)$

Solution:

Ans. (6)

$S_n = (1 + 2T_n)(1 - T_n) \Rightarrow T_1 = (1 + 2T_1)(1 - T_1)$

$T_1 = 1 - T_1 + 2T_1 - 2T_1^2 \Rightarrow 2T_1^2 = 1 \Rightarrow T_1 = \frac{1}{\sqrt{2}}$

$S_2 = T_1 + T_2 = (1 + 2T_2)(1 - T_2) \Rightarrow T_1 + T_2 = 1 - T_2 + 2T_2 - 2T_2^2$

$T_1 = 1 - 2T_2^2 \Rightarrow 2T_2^2 = 1 - \frac{1}{\sqrt{2}}$

$T_2^2 = \frac{\sqrt{2}-1}{2\sqrt{2}} \Rightarrow T_2^2 = \frac{2-\sqrt{2}}{4} \Rightarrow a = 4, b = 2 \Rightarrow a + b = 6$

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